

Artificial Intelligence in Speech-Language Pathology and Dysphagia: A Review From Latin American Perspective and Pilot Test of LLMs for Rehabilitation Planning[☆]

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Summary: Artificial Intelligence (AI) is transforming speech-language pathology (SLP) and dysphagia management, offering innovative solutions for assessment, diagnosis, and rehabilitation. This narrative review examines AI applications in these fields from 2014 to 2024, with particular focus on implementation challenges in Latin America. We analyze key AI technologies—including deep learning, machine learning algorithms, and natural language processing—that have demonstrated high accuracy in detecting voice disorders, analyzing swallowing function, and supporting personalized rehabilitation. The review identifies three primary domains of AI application: diagnostic tools with improved sensitivity for speech-language disorders, rehabilitation technologies that enable customized therapy, and telehealth platforms that expand access to specialized care in underserved regions. However, significant barriers persist, particularly in Latin America, where limited infrastructure, insufficient linguistic adaptation, and scarce regional datasets hamper widespread implementation. Our pilot study evaluating commercially available large language models for rehabilitation planning demonstrates their potential utility in generating structured therapy activities, especially in resource-constrained settings. While AI shows promise in enhancing clinical workflows and expanding service delivery, the evidence suggests that current applications remain predominantly focused on diagnosis rather than integrated rehabilitation. This review highlights the need for culturally and linguistically adapted AI models, expanded regional research collaborations, and regulatory frameworks that ensure ethical AI integration into SLP and dysphagia care, positioning these technologies as complementary tools that enhance rather than replace clinical expertise.

Key Words: Artificial intelligence—Speech-language pathology—Rehabilitation—Machine learning—Dysphagia.

INTRODUCTION

Disorders affecting speech, language, and swallowing, such as dysphagia and voice pathologies, represent a significant public health burden, particularly among older adults and individuals with neurological conditions. The global prevalence of oropharyngeal dysphagia ranges from 2.3% to 16% in the general population, reaching up to 52.7% in nursing home residents.^{1–3} Pediatric dysphagia is also increasing, particularly in children with neurodevelopmental disorders, preterm births, or aerodigestive conditions.⁴ Voice disorders, including dysphonia, affect approximately 16.9% of the population at some point in life, with higher rates among professionals who rely heavily on vocal function, such as teachers and singers.^{5–8} These conditions significantly impact quality of life, leading to nutritional deficiencies, aspiration pneumonia, and social isolation. However, there is no comprehensive estimation of these disorders in Latin America. These conditions can be caused

by various factors, including neurodevelopmental disorders, neurological diseases, and head and neck cancer, among others.

Artificial intelligence (AI) has emerged as a transformative tool in medicine, particularly in speech-language pathology (SLP) and dysphagia management. AI refers to the development of computational models that mimic cognitive functions such as learning, reasoning, and decision-making.^{9,10} In medicine, AI spans a broad spectrum of techniques, ranging from rule-based systems and traditional machine learning (ML) approaches to more advanced deep learning (DL) and generative AI models.^{11,12} While traditional statistical methods focus on inferential analysis and structured datasets, ML techniques can process large, unstructured, and multimodal data such as voice recordings, acoustic waveforms, and real-time speech monitoring.¹³

ML methods like support vector machines, random forests, and Gaussian mixture models have been widely used for pattern recognition in speech disorders, but their effectiveness is limited by their reliance on handcrafted feature extraction.¹⁴ In the field of SLP, AI has been applied to develop automated assessment and diagnostic tools, as well as to implement personalized rehabilitation strategies.^{15–18} Various studies have demonstrated the utility of AI in detecting and classifying voice disorders,^{19–21} evaluating dysphagia,^{15,22} and rehabilitating language in patients with aphasia.^{23,24} These applications have shown

Accepted for publication April 10, 2025.

[☆] This study did not have any financial source or support.

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Journal of Voice, Vol xx, No xx, pp. xxx–xxx

0892-1997

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<https://doi.org/10.1016/j.jvoice.2025.04.010>

promising results in terms of diagnostic accuracy, resource efficiency, and improvement in patient clinical outcomes.

The success of AI in speech and swallowing rehabilitation relies heavily on Big Data. Large-scale datasets such as AphasiaBank and PhonBank have been instrumental in training AI models for ASR and speech disorder classification.²⁵ Furthermore, wearable sensors, smartphone-based speech tracking, and AI-assisted telemedicine platforms are expanding access to real-time speech and swallowing disorder monitoring.^{26–28} However, the implementation of these technologies remains challenging due to technological barriers, limited access to high-quality training datasets, and variability in clinical protocols across different settings.^{29–31}

Despite the growing body of research on AI in voice and swallowing disorders, most studies originate from high-income countries, with limited data on its applicability in Latin America (LATAM). The region faces unique challenges, including healthcare disparities, lack of access to AI-powered diagnostic tools, and linguistic diversity, which complicates the development of speech models suited to regional populations.⁴ Additionally, bias in AI training datasets remains a concern, as many models are developed using data from monolingual English speakers, limiting their generalizability to multilingual and dialectally diverse populations.^{32,33} Addressing this gap, this review aims to:

1. Critically examine the state of AI applications in SLP and dysphagia, with focus in LATAM setting.
2. Assess the feasibility of integrating large language models (LLMs) into rehabilitation planning.
3. Present a pilot test evaluating the potential of LLMs for personalized therapy recommendations.

METHODS

Study design

This study follows a narrative review approach, aiming to synthesize and critically analyze the existing literature on the application of AI in SLP and dysphagia rehabilitation, with a particular emphasis on its implementation in LATAM. Given the exploratory nature of this review, a broad and flexible search strategy was employed to capture the diversity of available research.

Literature search strategy

A literature search was conducted in PubMed, Web of Science (WoS), and Scientific Electronic Library Online (SciELO) to identify relevant studies published between January 2014 and March 2024. The search terms included variations of AI, DL, ML, LLMs, speech therapy, voice pathology, swallow therapy, and dysphagia. To ensure coverage of both global developments and regional perspectives, the search was restricted to articles published in English, Spanish, and Portuguese, allowing for the inclusion of studies specifically addressing AI applications in Latin American contexts.

A total of 164 abstracts were initially screened from PubMed, followed by 108 additional abstracts retrieved from WoS, and two abstracts identified in SciELO. Given the specific focus on LATAM and the need for a comprehensive representation of the field, complementary searches were conducted in Google Scholar. The final sample considered 40 articles.

Study selection

The selection process involved an independent review of titles and abstracts by two researchers, who assessed the relevance of each study based on its focus on AI applications in SLP or dysphagia therapy, including diagnostic tools, automated assessment, and rehabilitation planning. Studies were included if they described AI, ML, DL, natural language processing (NLP), or LLMs as part of their methodology and provided insights into clinical or research applications of these technologies. Given the narrative nature of this review, article selection was conducted qualitatively, prioritizing diversity in methodologies, findings, and perspectives. Studies covering similar topics but contributing distinct insights, such as different AI models or clinical populations, were included to provide an overview of the field.

Data extraction and analysis

A qualitative thematic synthesis was applied to categorize and analyze findings from the selected studies. The main themes explored included AI-assisted diagnostics, treatment personalization, automation of speech and swallowing assessment, and barriers to implementation in Latin America. As this review was intended to capture the breadth of existing research rather than critically appraise individual study rigor, no formal quality assessment of the articles was performed.

RESULTS

AI applications in SLP

AI has played a transformative role in speech-language therapy (SLT), particularly in early diagnosis, personalized therapy, and patient monitoring. The integration of ML, DL, NLP, and generative AI models have significantly enhanced the accuracy and efficiency of speech disorder assessment and rehabilitation.^{34,35} Current AI applications in SLT can be categorized into three key areas: (1) AI-enhanced diagnostic tools, (2) AI-driven rehabilitation and personalized therapy, and (3) Emerging innovations in AI-powered speech therapy (Figure 1).

AI-powered diagnostic tools in SLT

The development of AI-based assessment tools has allowed for objective, real-time speech analysis, reducing the subjectivity and time constraints associated with traditional SLT evaluations. ASR systems and DL acoustic models are now being used to detect subtle voice and articulation

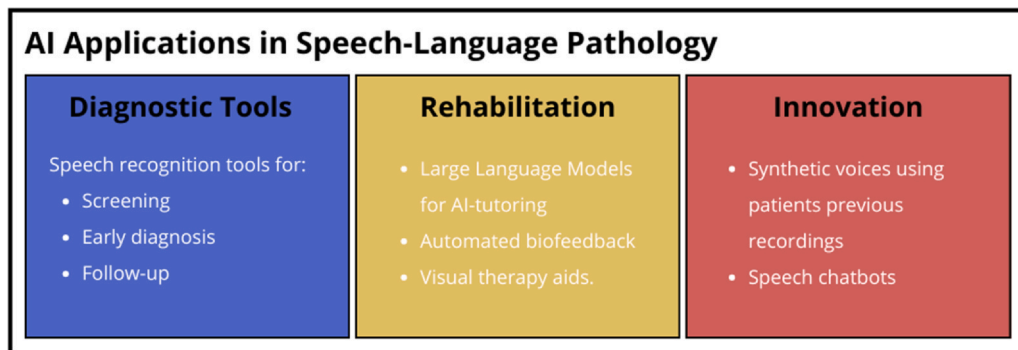


FIGURE 1. Key applications of artificial intelligence in speech-language pathology. This figure illustrates the three principal domains of AI applications in speech-language pathology: diagnostic tools (blue) that facilitate screening, early diagnosis, and follow-up through speech recognition technologies; rehabilitation systems (yellow) that incorporate large language models for AI-tutoring, automated biofeedback, and visual therapy aids; and innovative solutions (red) that enhance patient care through synthetic voice reconstruction and conversational speech chatbots (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

abnormalities, improving early diagnosis of conditions such as aphasia, dysarthria, and stuttering.^{23,24,36–39}

DL-based speech processing algorithms have demonstrated high accuracy rates in detecting phonetic abnormalities in patients with neurogenic speech disorders, for example in Parkinson's disease dysarthria.^{34,40–42} Furthermore, AI-powered phonetic assessment tools are now capable of analyzing prosody, articulation errors, and phonological patterns, making them potential screening for speech impairments in children and adults.^{43–45}

AI-driven rehabilitation and personalized therapy

AI can revolute speech rehabilitation by enabling adaptive, real-time therapy plans tailored to individual patients.³⁴ AI-driven intelligent tutoring systems leverage NLP and ML to provide personalized instructions and feedback for treating speech disorders in children, increasing motivation and engagement, which may lead to improved rehabilitation outcomes.^{36,46,47}

AI-powered biofeedback systems for voice disorder treatment utilize signal processing and real-time auditory or visual feedback to improve therapy effectiveness and accelerate recovery.⁴⁸ By analyzing voice quality and therapy progress, these systems offer adaptive, data-driven interventions, allowing that rehabilitation remains tailored to individual needs. AI-generated speech imaging tools, including AI-based spectrogram analysis and DALL-E-generated visual therapy aids, have also been introduced to support aphasia rehabilitation.⁴⁹

Emerging innovations in AI-powered speech therapy

Advances in DL-based voice cloning have enabled the creation of personalized synthetic voices for individuals who have lost their natural voice.⁵⁰ AI-driven speech chatbots,⁵¹ provide automated conversational exercises for

fluency training and articulation practice that can be extrapolated to simulate real-life interactions, thereby enhancing patient engagement and motivation during therapeutic interventions.

Furthermore, AI-powered remote automated diagnostic systems facilitate the assessment of symptom severity in patients with aphasia, enabling their participation in web-based speech therapy platforms.⁵² Similarly, AI systems support telemonitoring of symptom progression in Parkinson's disease patients.⁵³ These platforms can be particularly beneficial for rural and low-resource settings, where access to trained speech-language pathologists (SLPs) remains limited, while simultaneously improving adherence and therapeutic outcomes in telerehabilitation contexts.⁵⁴

AI applications in swallow therapy and dysphagia management

Dysphagia is a highly prevalent disorder that affects a wide range of populations, including stroke survivors, patients with neurodegenerative diseases, and older adults. AI-based approaches have the potential to transform dysphagia management by enhancing diagnostic accuracy, optimizing rehabilitation strategies, and personalizing treatment interventions.^{55–58} The role of AI in swallowing therapy can be categorized into three key domains: (1) AI-assisted dysphagia screening and assessment and (2) AI-powered swallowing rehabilitation (Figure 2).

AI-assisted dysphagia screening and assessment

Accurate assessment of dysphagia is crucial for guiding treatment and preventing aspiration pneumonia, malnutrition, and dehydration. Traditionally, videofluoroscopic swallow studies (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES) have been the gold standard diagnostic tools. However, these methods require

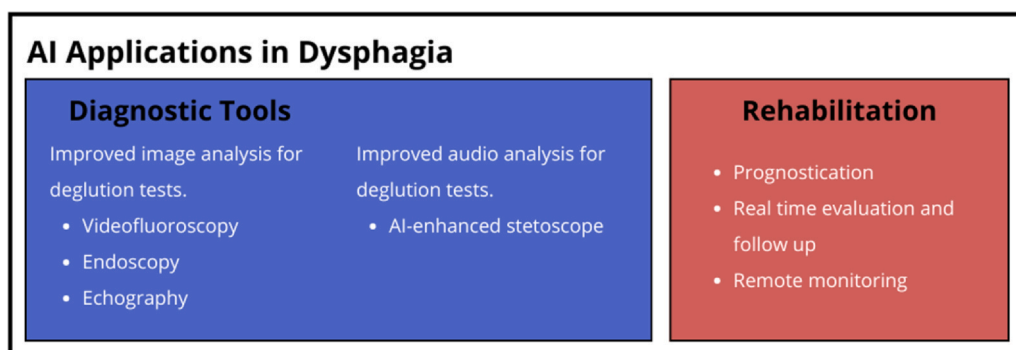


FIGURE 2. Applications of artificial intelligence (AI) in dysphagia management. Key domains of AI applications in dysphagia: diagnostic tools (blue) encompassing improved image analysis techniques for videofluoroscopy, endoscopy, and echography, alongside enhanced audio analysis using AI-enhanced stethoscopes; and rehabilitation approaches (red) that facilitate prognostication, real-time evaluation with follow-up capabilities, and remote monitoring of swallowing function (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

specialized personnel and are time-consuming, making AI-powered alternatives highly valuable.^{15,59,60}

DL models have demonstrated high accuracy in analyzing VFSS images, allowing automated detection of pharyngeal residue, penetration-aspiration events, and abnormal swallowing biomechanics,^{58,60,61} as well as supporting endoscopic expert evaluation.⁶¹ AI-driven motion tracking systems have been used to quantify hyoid bone displacement and laryngeal elevation in echography, providing objective kinematic data to clinicians.⁶² These approaches have shown comparable or superior accuracy to human evaluations, reducing subjectivity in interpretation.¹⁵

Additionally, ML algorithms trained on swallowing acoustics have been introduced as a non-invasive alternative to VFSS and FEES. AI models can analyze swallowing sounds recorded via electronic stethoscopes, identifying abnormal bolus flow, aspiration risk, and dysphagia severity.⁶³ These methods offer a bedside-friendly solution for real-time dysphagia assessment, particularly in neurological patients and Intensive Care Unit (ICU) settings.

AI-powered swallowing rehabilitation

Traditional dysphagia therapy employs clinician-supervised interventions including the effortful swallow, Mendelsohn maneuver, and neuromuscular electrical stimulation. AI applications in this domain have emerged in two primary capacities. First, ML models have demonstrated the ability to identify dysphagia severity and predict recovery outcomes six months post-stroke,⁶⁴ providing quantitative data for clinical prognostication. Second, AI systems have been developed to deliver real-time, data-driven rehabilitation tools that modify exercise parameters based on objective measurements of patient performance,⁶⁵ as well as dynamic tools to analyze safe and unsafe swallowing.⁶⁶

AI applications have been developed for dysphagia rehabilitation through remote monitoring systems that utilize inertial sensors and ML algorithms, enabling therapists to track patients' swallowing function in real time and make

personalized adjustments to intervention strategies.^{65,66} AI can also enhance dysphagia rehabilitation exercises by adapting them to each patient's specific capabilities and progress, improving treatment precision and accessibility.

Latin American experience in the use of AI-supported rehabilitation therapy for speech-language disorders and dysphagia

AI has enormous potential to transform healthcare accessibility in low- and middle-income countries, particularly in Latin America, a region with significant disparities in healthcare access.⁶⁷ While AI-driven innovations have been widely explored in high-resource healthcare systems, their implementation in LATAM remains limited, with most developments occurring within university-led research projects rather than widespread clinical adoption. Among the primary barriers to AI implementation are technological infrastructure gaps, insufficient funding for AI research, and the scarcity of large, regionally representative datasets required to develop linguistically and culturally adapted AI models.^{67,68}

In the field of SLT and dysphagia rehabilitation, AI-related experiences in LATAM are still in the early stages of development, primarily within academic research projects focused on four key areas^{69–74} (Figure 3):

1. Diagnostic evaluation and rehabilitation of dysphagia.
2. Professional training in AI-assisted technologies.
3. Automatic speech recognition (ASR) systems for therapy applications.
4. Mobile-based therapeutic support tools.

Despite these developments, AI-driven SLT and dysphagia care have yet to transition from research settings to widespread clinical practice in Latin America, highlighting the need for stronger research collaborations, regulatory frameworks, and investment in AI infrastructure.

The current limitations of AI deployment in SLT across the region reinforce the importance of developing cost-



FIGURE 3. Speech and language therapy AI-development in Latin America by Country. Areas of AI development in speech and language therapy across four Latin American countries, indicating the published experiences from Colombia (red coded), Argentina (blue coded), Mexico (green coded), and Perú (purple coded). AI, Artificial Intelligence (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

effective, accessible AI-driven solutions that do not require extensive computational resources. The use of simpler AI models, such as LLMs, represents a viable strategy for overcoming economic and infrastructural constraints. LLMs could provide automated therapy recommendations, adaptive rehabilitation planning, and AI-generated speech exercises, supporting clinicians in resource-limited environments. However, to ensure effective and ethical integration of AI into Latin American healthcare systems, further efforts are needed to develop culturally tailored AI models, expand training programs for healthcare professionals, and establish robust regulatory frameworks that support AI-based rehabilitation innovations in the region.

Pilot test of LLMs for rehabilitation planning

The pilot experience aimed to evaluate the feasibility of LLMs for generating rehabilitation activities in nursing home settings, particularly for patients requiring cognitive

maintenance and functional rehabilitation. The study was conducted in two phases, assessing both the quality and applicability of AI-generated rehabilitation exercises.

Phase 1: AI-based activity generation and initial testing

The initial step involved utilizing historical activity databases from two nursing homes, containing rehabilitation and cognitive stimulation exercises tailored to the needs of institutionalized patients. Given the necessity for continuous activity generation to maintain functional engagement among residents, AI models were tested for their ability to autonomously create new rehabilitation exercises based on existing therapeutic frameworks.

The evaluation included three state-of-the-art LLMs available at the time of the study:

- ChatGPT (OpenAI).
- Claude.AI (Anthropic).
- Gemini (Google).

The AI models were systematically tested over 10 consecutive days, generating sets of 10 rehabilitation activities per request under two conditions:

- Without historical activity support (pure LLM-based generation).
- With historical activity support (for models capable of incorporating reference data, ie, Claude.AI and ChatGPT).

The output from each AI model was assessed by two independent SLPs specialized in rehabilitation for elderly patients. The evaluation criteria focused on:

- Activity relevance and suitability for the target population.
- The necessity of modifications before implementation.
- Content diversity and non-repetitiveness.

Phase 2: Expert evaluation and practical implementation

All LLMs successfully generated rehabilitation activities, with notable improvements when provided with historical activity datasets. Experts found this enhancement significant but not crucial, as LLMs still required prompt refinement to ensure alignment with specific rehabilitation needs.

Regarding modifications needed before implementation, once the prompt structure was optimized for rehabilitation purposes, AI-generated activities required minimal adjustments. However, repetition of activities became evident over time, necessitating additional prompts to generate unique content as the test progressed. Claude.AI (with historical support) achieved the highest qualitative performance, followed by ChatGPT (with historical support).

Upon confirming the feasibility of AI-assisted activity generation, the AI-generated rehabilitation exercises were integrated into the pool of available rehabilitation activities within the nursing homes. No significant complaints were reported by SLPs utilizing these activities, nor by patients participating in therapy sessions.

Challenges and limitations of the pilot test

Several key limitations emerged during the proof-of-concept study.

1. **Variability in AI Performance Across Model Versions:** Differences in LLM performance across versions and between free and paid subscriptions highlighted the dynamic nature of AI capabilities, suggesting that future iterations of these models may continue to improve.
2. **Prompt Engineering for Rehabilitation Contexts:** Identifying the optimal prompt structure to generate clinically appropriate activities required three pilot iterations before achieving reliable outputs.
3. **Cultural and Regional Context Issues:** AI models struggled with country-specific contextual references, frequently misinterpreting local elements by substituting them with broader Latin American cultural references.

Future directions

A formal study evaluating the impact of AI-generated rehabilitation exercises is pending, requiring ethics committee approval and structured research protocols. However, from a conceptual and qualitative perspective, the expert assessment suggests that LLMs can serve as effective tools for generating rehabilitation activities, provided that prompting strategies are optimized to clarify therapeutic goals.

DISCUSSION

This narrative review explored the current landscape of AI in SLT and dysphagia rehabilitation, analyzing its applications in diagnosis, therapy planning, and intervention. AI-driven tools have demonstrated significant advances in improving diagnostic precision, optimizing therapy personalization, and enabling automated rehabilitation strategies. The application of ML, DL, and LLMs has provided innovative approaches to voice disorder detection, speech recognition, and dysphagia assessment, with AI-assisted videofluoroscopy (VFSS) and acoustic swallow tracking systems improving clinical workflows.⁶³ However, despite these advancements, the clinical adoption of AI remains fragmented. While research raises the potential superiority of AI-based assessments in terms of sensitivity and efficiency,¹⁵ their real-world implementation is still limited due to challenges in integration, regulatory frameworks, and clinician acceptance. Among the most promising areas of

future development is the use of LLMs in rehabilitation planning, particularly in generating structured therapy exercises and adapting treatment recommendations to individual patient needs.^{49,75}

Gaps persist in the literature, particularly in the integration of AI-generated tools to rehabilitation protocols, still depending almost exclusively in human-designed therapy plans. Although some studies propose that AI models can create functional therapy exercises,^{23,76} their long-term impact on treatment adherence and patient engagement remains unclear. Additionally, there is a lack of large-scale clinical validation for AI-driven rehabilitation tools, as most studies have been conducted in controlled experimental settings rather than routine clinical practice. One of the most pressing concerns is the need for more transparent and explainable AI (XAI) models, as current DL-based systems operate as "black boxes," making it difficult for clinicians to interpret and trust AI-generated recommendations.^{75,77}

In LATAM, AI adoption in SLT and dysphagia rehabilitation remains slow due to infrastructure limitations, economic constraints, and the lack of locally trained AI models. The limited availability of SLPs in rural and underserved areas has created a need for scalable and cost-effective rehabilitation solutions, and AI-driven teletherapy could play a key role in expanding access to services.^{78,79} However, the successful implementation of AI in LATAM requires the development of models trained on Spanish and Portuguese data, ensuring linguistic and cultural adaptability. Given the challenges of implementing high-complexity AI solutions in LATAM healthcare systems, the use of simpler, more accessible AI tools—such as LLMs—represents a practical alternative. Our pilot study showed that LLMs can generate structured rehabilitation activities, reducing the cognitive load on therapists and increasing efficiency in therapy planning. While AI-generated exercises do not replace clinical expertise, they can complement rehabilitation strategies by automating aspects of activity development, particularly in settings with limited professional resources. The ability of LLMs to function with minimal computational infrastructure also makes them a viable solution for resource-limited healthcare environments.⁸⁰

Despite its potential, AI in rehabilitation faces multiple limitations, including bias in training datasets, the evolving nature of AI models, and the need for ongoing clinician oversight. Many AI models are trained predominantly on English-language datasets, making their applicability to LATAM populations less reliable.⁸¹ This highlights the urgent need for regional datasets to improve model accuracy and generalizability.⁸² Another key challenge is the continuous evolution of LLMs and DL models,^{83,84} as performance may vary between versions, requiring constant reassessment of AI-generated outputs. Ethical and regulatory considerations must also be addressed to prevent over-reliance on AI at the expense of clinical judgment.^{85–87} As AI adoption increases, it is essential that SLPs receive appropriate training to integrate these

technologies effectively, ensuring that AI enhances rather than replaces human expertise in rehabilitation.¹⁵ Additionally, the cost of AI implementation remains a concern, particularly in public healthcare systems where funding for technological advancements is limited.

CONCLUSION

AI could redefine rehabilitation by enabling more precise diagnoses, optimizing therapy workflows, and expanding access to rehabilitation services, particularly in regions with limited healthcare infrastructure. While significant progress has been made in AI-driven SLT and dysphagia diagnosis enhancement, substantial gaps remain in rehabilitation integration, clinical validation, and contextual adaptation for diverse patient populations.

The potential for AI-driven rehabilitation in LATAM depends on the ability to develop linguistically and culturally adapted AI tools while ensuring accessibility and regulatory support. Our findings suggest that LLMs offer a viable starting point for integrating AI into rehabilitation planning, particularly in resource-limited settings. However, further research is needed to assess long-term outcomes, refine model accuracy, and develop ethical frameworks that promote safe and effective AI integration in clinical practice.

Ethical Approval

This study do not require ethics committee approval.

Author Agreement Statement

All authors have approved the final article, agreed to be accountable for all aspects of the work and acknowledge that all those entitled to authorship are listed as authors.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andres Giglio Jiménez: Received grants or contracts from Pfizer within the last 36 months about the Infection diseases in immunocompromised critical patients course funding, to institution.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Paperpal to improve the redaction of this paper. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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